

Worksheet 3

Problem 1. Polynomial Spaces & Linear Independence

Consider the vector space $V = \mathcal{P}_3(\mathbb{R})$ of polynomials with real coefficients of degree at most 3. Let $S = \{p_1(x), p_2(x), p_3(x), p_4(x)\}$ be a subset of V , where:

$$p_1(x) = 1 + 2x - x^2$$

$$p_2(x) = 3 + x^2 + x^3$$

$$p_3(x) = 2 - 2x + 2x^2 + x^3$$

$$p_4(x) = 4 + 4x - x^2 + x^3$$

- (a) Determine whether S is linearly independent or linearly dependent.
- (b) Find a basis for $\text{Span}(S)$ and state its dimension.
- (c) Extend the basis found in part (b) to a basis for all of $\mathcal{P}_3(\mathbb{R})$.

Problem 2. Subspaces & Dimension

Let W be the subset of $\mathcal{P}_2(\mathbb{R})$ defined by:

$$W = \{p(x) \in \mathcal{P}_2(\mathbb{R}) \mid p(1) = 0 \text{ and } p'(0) = 0\}$$

- (a) Prove that W is a subspace of $\mathcal{P}_2(\mathbb{R})$.
- (b) Find a basis for W and determine $\dim(W)$.
- (c) Construct a subspace $U \subset \mathcal{P}_2(\mathbb{R})$ such that $\mathcal{P}_2(\mathbb{R}) = W \oplus U$ (a direct sum).

Problem 3. Linear Maps on Polynomial Spaces

Define a transformation $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ by:

$$T(p(x)) = x \cdot p'(x) + p''(x)$$

- (a) Prove that T is a linear map.
- (b) Find the matrix representation $[T]_B$ of T relative to the standard basis $B = \{1, x, x^2\}$.
- (c) Determine the nullity ($\dim(\ker(T))$) and rank ($\dim(\text{im}(T))$), and state whether T is invertible.

Extra Prob.

i.) Show $\dim(\text{Null}(ST)) \leq \dim(\text{Null}(S)) + \dim(\text{Null}(T))$.

$\rightarrow$ Let $\{v_1, \dots, v_k\}$ be a basis for $\text{Null}(T)$ & $\{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ for $\text{Null}(ST)$. Then, $T(v_{k+1}), \dots, T(v_n) \in \text{Null}(S)$ must be lin. indep. $\Rightarrow \leq \dim(\text{Null}(S))$ many.

1.)

A.) These are linearly indep. iff.

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 2 & 0 & 2 & 4 \\ -1 & 1 & 2 & -1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

has trivial Null space. By row ops,

$$\sim \begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & -6 & -6 & -8 \\ 0 & 4 & 4 & 3 \\ 0 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So they are dependent.

B.) Pivots in col. 1, 2, 4 $\Rightarrow$ basis p_1, p_2, p_4 (dim 3).

C.) $\{p_1, p_2, p_4, x^2\}$

2.) If $p(x, q(x)) \in W$ and $d \in \mathbb{R}$,

A.) $(dp(x) + q(x))'(0) = dp'(0) + q'(0) = 0$

and $(dp + q)(1) = dp(1) + q(1) = 0$

So $dp + q \in W$. Since $0 \in W$, this is a subspace.

B.) If $p(x) = a_0 + a_1x + a_2x^2$ is in W , $p(1) = a_0 + a_1 + a_2 = 0$

and $p'(0) = a_1 + 0 = 0$, so $p(x) = a_0 - a_0x^2$.

Hence, W has basis $\{1 - x^2\}$. (1D).

C.) $V = \text{span}\{1, x\}$.

3.)

$$A.) T(\alpha p + q) = \alpha(\alpha p' + q') + \alpha p'' + q''$$

$$\alpha [x(p') + p''] + [xq' + q''] = \alpha T(p) + T(q)$$

So T is linear.

$$B.) T(1) = 0$$

$$T(x) = x$$

$$T(x^2) = 2 + 2x^2$$

$$\rightarrow \begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$C.) \text{ Nullity} \rightarrow 1$$

$$\text{Rank} \rightarrow 2$$

Not invertible.

Friday 9/18 Disc Notes

Concepts

- Dimension - length of basis

- 1 length b/c deletion lemma $\Rightarrow$

length of L.I. list $\leq$ length of spanning list.

$$- \dim(U+V) = \dim(U) + \dim(V) - \dim(U \cap V)$$

~ Build up 2 Bases

- Linear Maps. - for V, W v.s. over $\mathbb{F}$, $T: V \rightarrow W$ is linear

$$\text{if } T(\alpha v + u) = \alpha T(v) + T(u) \quad \forall \alpha \in \mathbb{F}, v, u \in V$$

- The map commutes with vector space ops.

$$\text{e.g.) } T: \mathbb{P} \rightarrow \mathbb{R}^3$$

$$\text{by } T(p(x)) = (p(0), p'(0), \int_0^1 p(x) dx)$$

$$\text{Null}(T) = \{v \in V \mid T(v) = 0\} \rightarrow T \text{ injective iff. } \text{Null}(T) = \{0\}$$

$$\text{Range}(T) = \{w \in W \mid w = T(v) \text{ for some } v \in V\} \rightarrow T \text{ surjective iff. } \text{Range}(T) = W.$$

Thm: If V is finite-dimensional,

$$\dim(\text{Null}(T)) + \dim(\text{Range}(T)) = \dim(V).$$

$$\text{e.g.) } \pi_1: \mathbb{R}^3 \rightarrow \mathbb{R} \quad \text{by } \pi_1(x, y, z) = x.$$